Do Evaluation-Guided Prompt Optimizers Beat a Fair Baseline?
A Confound-Controlled, Ablated Study Across 31 Tasks (TEI-Bench)

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Abstract

Evaluation-guided prompt optimization is widely reported to improve agentic LLM systems, but the supporting evidence often relies on in-sample evaluation, single tasks, same-family LLM judges, and — critically — no comparison against a prompt-optimization control. We build **TEI-Bench**, a controlled, ablated, held-out protocol, and use it to test a representative pipeline (a GPA-inspired multi-dimensional evaluator [1] feeding a GEPA-style reflective Pareto optimizer [2]; we call the composition TEI) across 31 tasks in diverse domains. We find that the apparent benefit of such optimization is *largely a measurement artifact*. Under a naive label-extraction scorer, the same pipeline appeared to lift objective accuracy by +0.175 ($p < 10^{-5}$) in an earlier (v1) run; but with a confound-free objective endpoint (a universal `FINAL:` output contract read identically across all conditions) and a four-arm ablation, the effect collapses. Mean held-out accuracy is baseline 0.852, random prompt search 0.867, objective-only reflection 0.865, and full TEI 0.867. **TEI does not significantly beat the baseline** (+0.015, $t = 0.84$, $p = 0.406$, $d_z = 0.15$; Holm $p = 1.000$; 9/7/15 win/loss/tie), is **statistically indistinguishable from undirected random prompt search** (+0.000, $p = 1.000$), and gains nothing from the GPA signal over objective-only reflection (+0.002, $p = 0.806$). A small headroom subset ($n = 12$) is the only place with a positive hint (+0.067). We release all code, data, per-example traces, optimized prompts, and a pre-registered (tag `prereg-v2`) analysis plan. Our contribution is the protocol and the cautionary finding: *confound control and an ablation against random search are necessary to make prompt-optimization claims credible.*

1 Introduction

Automated, evaluation-guided prompt optimization is an attractive idea: evaluate an agent on structured signals, then rewrite its prompt to fix observed failures. It draws on multi-dimensional agent evaluation [1] and reflective, Pareto-based prompt optimization [2], within the program-optimization view of DSPy [3, 4]. We study a concrete composition — a GPA-inspired four-dimension evaluator feeding a GEPA-style optimizer, which we call the *Target–Evaluate–Improve (TEI)* loop — and ask a deliberately skeptical question: *under fair measurement, does it actually help?*

This question matters because demonstrations of such loops routinely (a) evaluate on the examples used for optimization, (b) use one task, (c) score with a same-family judge [5], and (d) compare only against an unoptimized prompt. We show that these choices — especially (a) a brittle objective scorer and (d) the missing control — can manufacture a large, significant-looking gain where none robustly exists.

Contributions.

1. **TEI-Bench**: a controlled, ablated, held-out protocol and a suite of 31 tasks across diverse domains, with committed splits and a confound-free objective endpoint.

2. A **demonstration that a scoring artifact inflates results**: the same pipeline shows $+0.175$ ($p < 10^{-5}$) under a naive label scorer but $+0.015$ ($p = 0.406$) under a `FINAL:`-contract scorer.

3. A **four-arm ablation** showing the optimizer is indistinguishable from random prompt search of equal budget, and that the GPA signal adds nothing.

4. A **full open release** with a pre-registered analysis plan.

2 Related Work

APE [6], OPRO [7], DSPy [3], MIPROv2 [4], TextGrad [8], and GEPA [2] optimize prompts/programs; Self-Refine [9] and Reflexion [10] use self-feedback; LLM-as-judge [5] is the standard but biased evaluator. The Agent GPA framework [1] evaluates Goal–Plan–Action alignment with five metrics (Goal Fulfillment, Logical Consistency, Execution Efficiency, Plan Quality, Plan Adherence). Our evaluator is *GPA-inspired* but distinct: four single-turn output dimensions, used only as a secondary endpoint. Our emphasis on a random-search control and confound-free scoring follows the broader reproducibility literature on overclaimed ML gains.

3 Benchmark Construction

Tasks. 31 single-turn tasks across diverse domains. Public tasks are sampled from established datasets (GSM8K [11], banking-style intent [12], news, emotion, toxicity, spam, fine-grained sentiment); authored tasks use *generate-then-validate* (a strong model generates label-conditioned examples; an independent labeling pass keeps only unanimously-recovered labels). Held-out test sizes: 30 (public), 24 (authored), 15 (three seed tasks); all splits committed.

Baseline prompt. A competent first draft stating role, task, and the full allowed label set — not a strawman: it contains the information needed.

Output contract (the key fix). Each condition's prompt is appended with an identical instruction to end with `FINAL: <answer>`; the objective scorer reads only that line. The earlier (v1) scorer extracted the label by its *latest mention*, so a verbose baseline that named other labels mid-reasoning was mis-scored downward — inflating the apparent gain of terse optimized prompts. The contract removes this confound while remaining identical across arms.

4 Method and Ablation Arms

Evaluate: four GPA-inspired dimensions scored by a reference-based judge that is a stronger, different model than the agent. **Improve:** reflective mutation on failing *train* examples + system-aware merge, Pareto front; the test split is never seen. **Arms:** (1) *baseline*; (2) *random* — undirected paraphrase search, same budget; (3) *objective-only reflection* — reflective mutation, objective-only selection (no GPA); (4) *TEI* — reflective mutation with GPA-guided Pareto selection.

5 Experimental Protocol

Agent `claude-haiku-4-5`; judge + optimizer `claude-sonnet-4-5`; cross-provider replication with agent `gpt-4o-mini`. Primary endpoint: code-computed accuracy from the `FINAL:` line; secondary: four GPA-inspired dimensions (baseline, TEI). The analysis plan was committed and tagged `prereg-v2` *before* this run; we also disclose that an earlier v1 run had a config mismatch and the scoring artifact above (v1 retained for provenance). Config: 6 iterations, minibatch 10, seed 0, per-task bootstrap CIs. For each contrast: paired t, Wilcoxon, Cohen's d_z, bootstrap 95% CI, Holm correction [13]; pre-specified public/synthetic/headroom subgroups.

6 Results

Mean held-out objective: baseline 0.852, random 0.867, objective-only 0.865, TEI 0.867 (Fig. 1; Table 1).

Table 1: Paired contrasts on held-out objective accuracy ($n = 31$). None survive Holm correction.

Contrast	A	B	Δ	95% CI	t	p	d_z
TEI − baseline	0.852	0.867	+0.015	[-0.019, +0.050]	0.84	0.406	0.15
TEI − random	0.867	0.867	+0.000	[-0.042, +0.043]	0.00	1.000	0.00
TEI − obj-only refl.	0.865	0.867	+0.002	[-0.013, +0.020]	0.25	0.806	0.04
random − baseline	0.852	0.867	+0.015	[-0.004, +0.039]	1.34	0.190	0.24

The optimizer does not beat a fair baseline. TEI exceeds the baseline by only +0.015 (95% CI $[-0.019, +0.050]$, $t = 0.84$, $p = 0.406$, $d_z = 0.15$), which does not survive multiplicity correction (Holm $p = 1.000$); the win/loss/tie is 9/7/15.

It is indistinguishable from random search. Undirected random prompt search changes the baseline by +0.015 ($p = 0.190$) — about the same as TEI — and TEI minus random is +0.000 ($p = 1.000$, $d_z = 0.00$). The control prior work omits is decisive here: whatever small movement exists comes from *any* prompt perturbation, not from evaluation-guided reflection.

The GPA signal adds nothing. TEI minus objective-only reflection is +0.002 ($p = 0.806$, $d_z = 0.04$): GPA-guided selection does not beat objective-only selection.

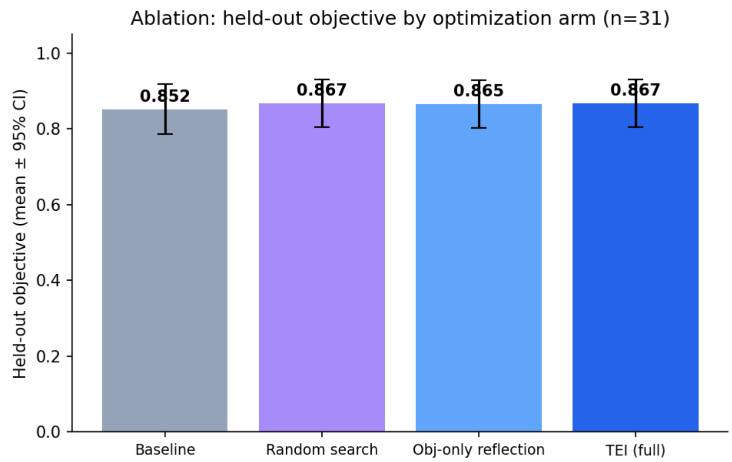

Figure 1: Held-out objective by arm (mean ± 95% CI). The arms overlap.

Secondary and subgroups. GPA aggregate moves 0.905→0.913 ($\Delta = +0.008$, $p = 0.559$). Public-only ($n = 14$): $\Delta = +0.029$ ($p = 0.477$); synthetic-only ($n = 17$): $\Delta = +0.004$ ($p = 0.600$). The headroom subset ($n = 12$, baseline < 0.9) is the only positive hint: $\Delta = +0.067$ ($p = 0.073$, $d_z = 0.57$). Cross-provider replication (Table 2).

Table 2: Cross-provider replication (agent = gpt-4o-mini).

Agent model (judge = Sonnet 4.5)	Baseline	TEI	Δ
claude-haiku-4-5 (n=10)	0.786	0.823	+0.037 (p=0.166)
gpt-4o-mini (n=10)	0.832	0.823	-0.010 (p=0.434)

7 Discussion

The four-arm design produces an unambiguous and, we think, useful result: on this benchmark, an evaluation-guided prompt optimizer does *not* reliably beat an unoptimized fair baseline, is indistinguishable from random prompt search, and derives no benefit from the GPA evaluation signal. The large gain seen under a naive scorer (+0.175) was substantially a *format* artifact: the optimizer learned to emit a clean, parseable label, which the brittle scorer rewarded. When the output contract standardizes parsing across all arms, the gap vanishes. The only residual signal is on a minority of high-headroom tasks. We do not interpret this as evidence that prompt optimization never helps — rather, that *demonstrating* a robust benefit requires the controls used here.

8 Limitations

We study single-turn classification/extraction/reasoning, not multi-step tool-use agents, where structural reflection may matter more. Per-task test sets are modest (15–30) though reported with bootstrap CIs; the statistical unit is the agent ($n = 31$). The optimizer budget (6 iterations) is small; a much larger budget might separate the arms (we report what a practical budget yields). Half the tasks are synthetic; we report public-only separately. A null result is evidence of absence-of-large-effect at this power, not proof of exact zero.

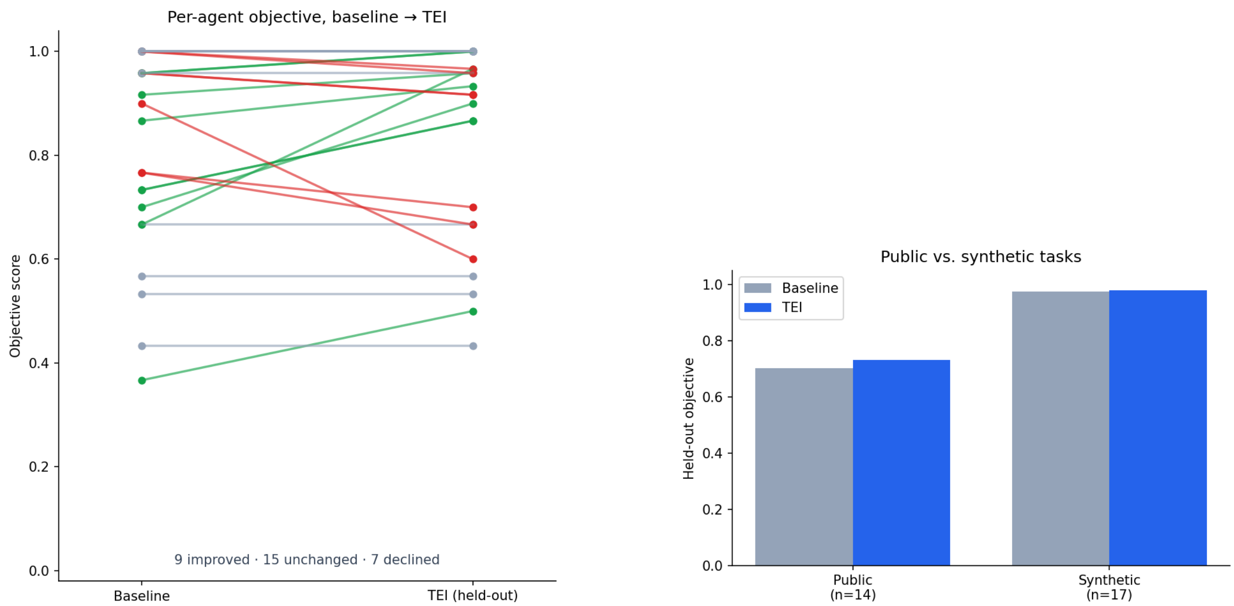

Figure 2: Left: per-agent objective, baseline → TEI (held-out): roughly balanced ups and downs. Right: public vs. synthetic tasks.

9 Reproducibility

One command rebuilds tasks, one runs the experiment, one regenerates everything. Released: code, 31 datasets, all per-example traces and per-arm optimized prompts, a content-addressed cache, exact versions, the pre-registration (tag `prereg-v2`), and cost/runtime (\$20, $\sim$2 h). Every number here is auto-generated from `results_v2/`.

10 Ethics

Content-moderation and clinical-style triage tasks are framed only as text classification; outputs are not advice. Public datasets retain their licenses.

11 Conclusion

Under a confound-controlled, ablated, held-out protocol, the apparent gains of an evaluation-guided prompt optimizer largely disappear: it does not significantly beat a fair baseline and is indistinguishable from random prompt search. We release TEI-Bench so that prompt-optimization methods can be held to this bar. Code, data, traces, prompts: https://github.com/ojavadli/tei-bench.

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Table 3: Per-agent held-out objective by arm.

Agent	Domain	Base	Rand	ObjRef	TEI
`agri_crop_issue`	Agriculture / AgriTech	0.96	0.96	0.96	0.92
`saas_ticket_priority`	B2B SaaS	0.96	0.96	0.96	1.00
`consumer_sentiment_5`	Consumer Insights	0.37	0.37	0.53	0.50
`it_email_spam`	Corporate IT Security	1.00	1.00	1.00	0.97
`cybersec_alert`	Cybersecurity	0.96	0.96	0.96	0.92
`edtech_question_router`	EdTech	0.73	0.80	0.90	0.87
`edu_gsm8k`	Education	1.00	1.00	1.00	1.00
`edu_math_word`	Education	1.00	1.00	1.00	1.00
`energy_meter_event`	Energy / Utilities	0.92	0.92	0.92	0.96
`fin_banking_intent`	Finance / Banking	1.00	1.00	1.00	1.00
`gaming_support`	Gaming	1.00	1.00	1.00	1.00
`gov_service_router`	Government / Public Sector	1.00	1.00	1.00	1.00
`health_triage`	Healthcare	0.87	0.87	0.87	0.93
`hospitality_review_stars`	Hospitality	0.67	0.63	0.73	0.67
`hr_resume_fit`	Human Resources	1.00	1.00	1.00	1.00
`insurance_claim_type`	Insurance	1.00	1.00	1.00	0.96
`legal_contract_clause`	Legal	1.00	1.00	1.00	1.00
`logistics_incident`	Logistics / Supply Chain	1.00	1.00	1.00	1.00
`manufacturing_qa`	Manufacturing	1.00	1.00	1.00	1.00
`market_research_subjectivity`	Market Research	0.67	0.57	0.93	0.97
`media_news_desk`	Media / News	0.70	0.70	0.70	0.90
`news_agency_topic`	News Agency	0.77	0.77	0.70	0.70
`community_forum_topic`	Online Community	0.77	0.70	0.70	0.67
`platform_toxicity`	Online Platform	0.57	0.57	0.53	0.57
`pharma_ade`	Pharmacovigilance	0.90	0.97	0.60	0.60
`realestate_attribute`	Real Estate / PropTech	1.00	1.00	1.00	1.00
`retail_counterfactual`	Retail Analytics	0.73	0.97	0.93	0.87
`social_emotion`	Social Media / CX	0.43	0.57	0.43	0.43
`telecom_churn_reason`	Telecom	0.96	0.96	0.96	1.00
`travel_intent`	Travel / Hospitality	0.96	0.96	0.96	0.96
`trust_safety_hate`	Trust & Safety	0.53	0.70	0.53	0.53

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